UNIT 6 · FUNCTIONS · 8TH GRADE MATH

Tables & Systems of Equations

ANSWER KEY

Guided Notes

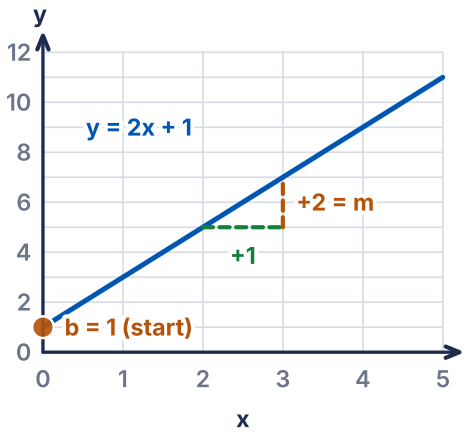
1 Writing Linear Equations from a Table

The Big Idea

Every linear equation can be written as **y = mx + b**. To go from a table to an equation, we need to find two things:

- **m** = the slope (the rate of change)
- **b** = the y-intercept (the y-value when **x = 0**)

$$y = mx + b$$



b = where the line starts on the y-axis. **m** = how much y changes each time x goes up 1.

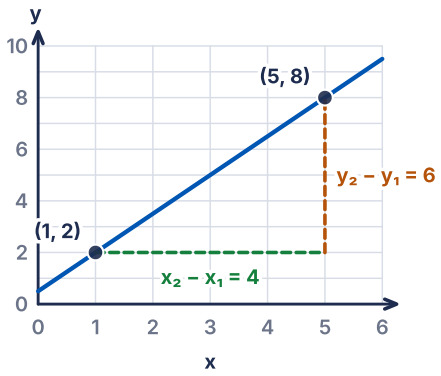
QUICK CHECK: IS THE TABLE LINEAR?

A table shows a linear relationship only when the y-values change by the *same amount* for equal steps in x. If x goes up by 1 each row and y jumps around, it's not linear.

Finding the Slope from a Table

$$m = (y_2 - y_1) \div (x_2 - x_1) = \text{change in } y \div \text{change in } x$$

Pick any two rows of the table. Subtract to get the change in y and the change in x. Divide.



m = 6 ÷ 4 = 1.5

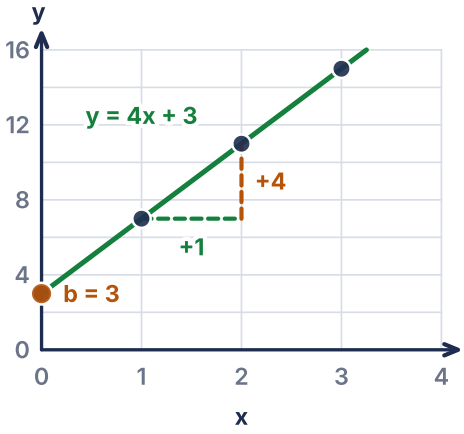
EXAMPLE 1 **Table includes x = 0**

x	y
0	3
1	7
2	11
3	15

1. **Find the slope.** From row 1 to row 2: y changes by `+4`, x changes by `+1`. So `m = 4`.
2. **Find the y-intercept.** When x = 0, y = 3. So **`b = 3`**.
3. **Write the equation:** `y = 4x + 3`

x	y	Δy ÷ Δx
0	3	
1	7	$\frac{4}{1} = 4$
2	11	$\frac{4}{1} = 4$
3	15	$\frac{4}{1} = 4$

+1, +4 between each pair of rows

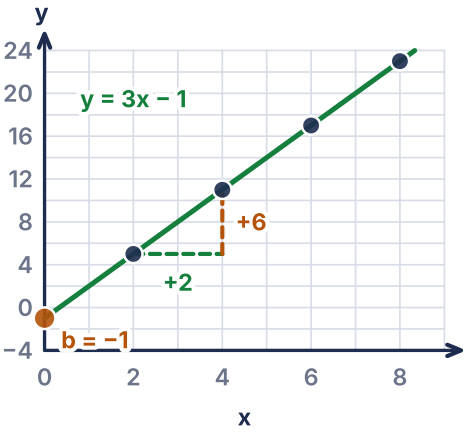


EXAMPLE 2 **Table does NOT include x = 0**

x	y
2	5
4	11
6	17
8	23

1. **Find the slope.** From (2, 5) to (4, 11): Δy = 6, Δx = 2, so $m = \frac{6}{2} = 3$.
2. **Work backward to find b.** Start at `(2, 5)`. Back up x by 1 means y drops by 3. So at `x = 1, y = 2`. At `x = 0, y = −1`. So `b = −1`.
3. **Write the equation:** `y = 3x − 1`
4. **Check.** Plug in x = 8: y = 3(8) − 1 = 23. ✓ Matches the table.

x	y	Δy ÷ Δx
0	−1	
		$\frac{3}{1} = 3$
1	2	
		$\frac{3}{1} = 3$
2	5	
		$\frac{6}{2} = 3$
4	11	
		$\frac{6}{2} = 3$
6	17	
		$\frac{6}{2} = 3$
8	23	

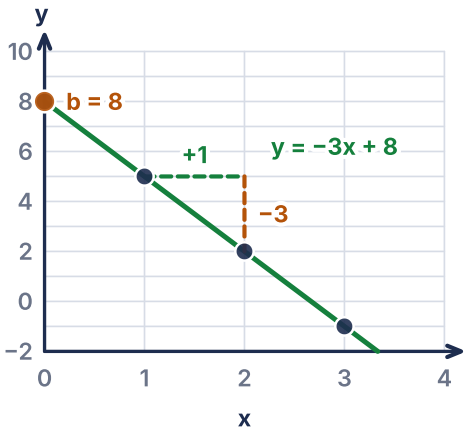


EXAMPLE 3 **Negative slope (y decreases as x increases)**

x	y
0	8
1	5
2	2
3	−1

1. **Find the slope.** From (0, 8) to (1, 5): Δy = `−3`, Δx = `+1`. So `m = −3`. (When y *decreases* as x increases, the slope is negative.)
2. **Find the y-intercept.** When x = 0, y = 8. So `b = 8`.
3. **Write the equation:** `y = −3x + 8`
4. **Check.** Plug in x = 3: y = −3(3) + 8 = −1. ✓ Matches the table.

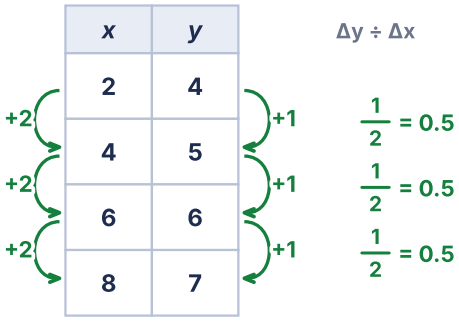
x	y	Δy ÷ Δx
0	8	
		$\frac{-3}{1} = -3$
1	5	
		$\frac{-3}{1} = -3$
2	2	
		$\frac{-3}{1} = -3$
3	−1	

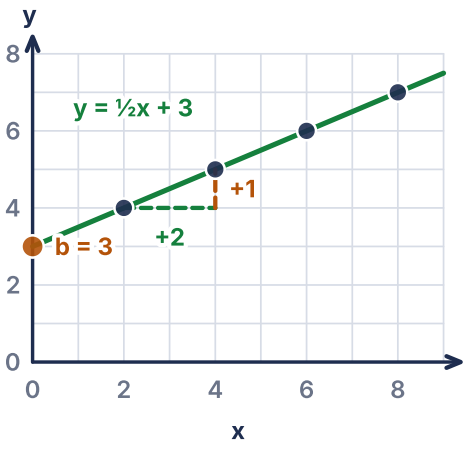


EXAMPLE 4 **Fractional slope**

x	y
2	4
4	5
6	6
8	7

1. **Find the slope.** From (2, 4) to (4, 5): Δy = `1`, Δx = `2`. So `m = 1/2`. (Don't round — keep it as a fraction.)
2. **Work backward to find b.** Start at `(2, 4)`. Back up x by 2 (to x = 0) means y drops by 1. So at `x = 0, y = 3`. So `b = 3`.
3. **Write the equation:** `y = (1/2)x + 3`
4. **Check.** Plug in x = 8: y = $(\frac{1}{2})(8) + 3 = 4 + 3 = 7$. ✓ Matches the table.





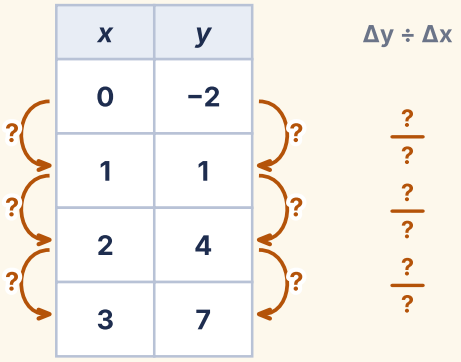
CHALLENGE: A SHORTCUT FOR FINDING B

Once you know *m*, pick **any** row (x, y) from the table and solve: `b = y − m·x`

Example 2 check: `b = 5 − 3(2) = −1`. ✓

▶ Try It · Table A

x	y
0	−2
1	1
2	4
3	7

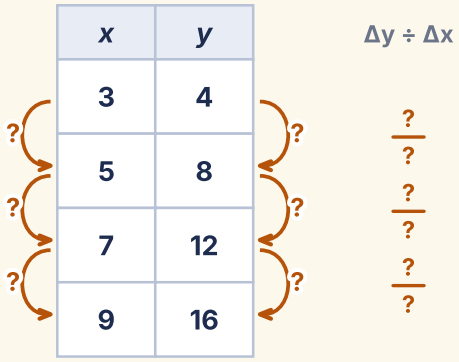


m = 3 **b =** −2

> **y = 3x − 2** · slope = 3 (y changes by +3 each row), y-intercept = −2 (when x=0)

▶ Try It · Table B (no x = 0)

x	y
3	4
5	8
7	12
9	16



No x = 0 row? After you find m, work backward up the table to x = 0.

m = 2 **b =** −2

> **y = 2x − 2**
> Slope: $\frac{\Delta y}{\Delta x} = \frac{4}{2} = 2$.
> y-intercept: work backward from (3, 4). Each step back of 1 in x, y drops by 2: (2, 2), (1, 0), (0, −2). So b = **−2**.

✏ Practice: Writing Equations from a Table

1. Score 3 What is the slope **m** of this table? Write just the number.

Answer: 2

x	y
0	5
1	7
2	9
3	11

How to do it: Each time x goes up 1, y goes up 2 (5 → 7 → 9 → 11). So m = 2 ÷ 1 = **2**.

2. Score 4 Write the equation for this table in the form y = mx + b.

Answer: y = −4x + 10

x	y
0	10
1	6
2	2
3	−2

How to do it: Slope: y changes by −4 each time x goes up 1, so m = −4. When x = 0, y = 10, so b = 10. Equation: **y = −4x + 10**.

3. Score 5 This table has no x = 0 row. Find **b**, the y-intercept. Write just the number.

Answer: b = −5

x	y
2	1
4	7
6	13
8	19

How to do it: m: y goes up 6 when x goes up 2, so m = 6 ÷ 2 = 3. Work backward from (2, 1): x goes back 2 to 0, so y drops 3 × 2 = 6. 1 − 6 = **−5**. Equation: y = 3x − 5.

2 Solving Systems by Graphing

What Is a System of Equations?

A **system of equations** is two or more equations that share the same variables. In 8th grade, we work with systems of **two linear equations** in two variables (x and y).

EQ 1 $y = 2x + 1$

EQ 2 $y = -x + 4$

x	y
0	1
1	3
2	5
3	7

EQ 1: y = 2x + 1

x	y
0	4
1	3
2	2
3	1

EQ 2: y = −x + 4

Both tables have the row x = 1, y = 3. That shared pair works in **both** equations.

Graphing Method: The Big Idea

Graph each line on the same coordinate plane. The point where the two lines **intersect** is the solution.

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solution of a system = intersection point (x, y)
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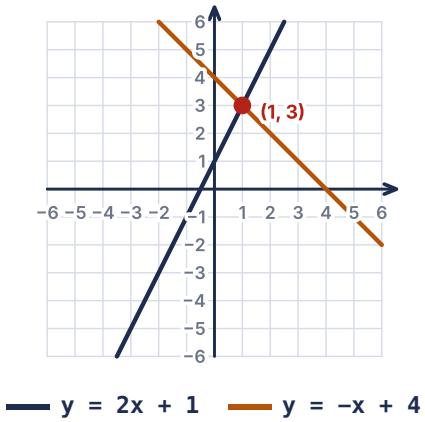
STEPS

1. Graph the first line (use slope and y-intercept, or a table).
2. Graph the second line on the same plane.
3. Find where they cross. That ordered pair is the solution.
4. Check by plugging the point into both equations.

EXAMPLE **Solve by graphing**

EQ 1 $y = 2x + 1$

EQ 2 $y = -x + 4$



1. **EQ 1:** slope 2, y-int 1. Graph from (0, 1): right 1, up 2 → (1, 3), (2, 5)...
2. **EQ 2:** slope −1, y-int 4. Graph from (0, 4): right 1, down 1 → (1, 3), (2, 2)...
3. **Read the intersection:** `(1, 3)`
4. **Check.** EQ 1: 3 = 2(1) + 1 = 3 ✓. EQ 2: 3 = −(1) + 4 = 3 ✓. Both true. **Solution: (1, 3)**.

Three Possible Outcomes

When you graph a system of two lines, one of three things will happen:

CASE 1 · ONE SOLUTION

The two lines **cross at exactly one point**. The system has **one solution** — that point. (Different slopes.)

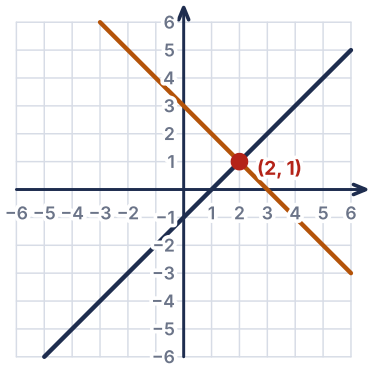
CASE 2 · NO SOLUTION

The two lines are **parallel**. They never cross. The system has **no solution**. (Same slope, different y-intercepts.)

CASE 3 · INFINITELY MANY SOLUTIONS

The two equations are the **same line**. Every point on the line works. **Infinitely many solutions**. (Same slope AND same y-intercept.)

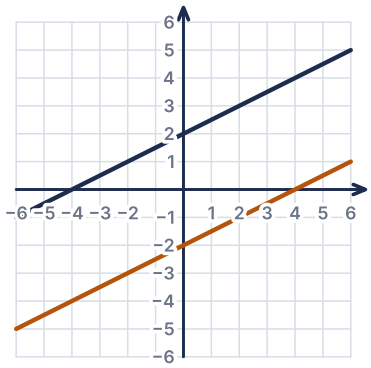
CASE 1 · ONE SOLUTION



They cross once

y = x − 1 y = −x + 3

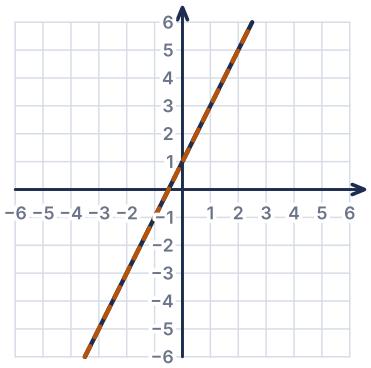
CASE 2 · NO SOLUTION



Parallel — never cross

y = ½x + 2 y = ½x − 2

CASE 3 · INFINITELY MANY



One line, drawn twice

y = 2x + 1 y = 2x + 1

Count the intersection points: **one** point → one solution, **zero** points → no solution, **every** point → infinitely many.

▶ Try It · Read the Graph

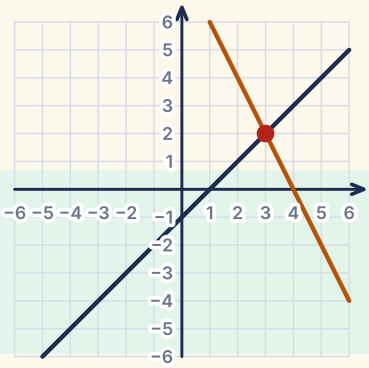
Read the graph: the two lines cross at the red point. Write the solution as an ordered pair.

Solution: (3 , 2)

Solution: (3, 2)

Check EQ 1: y = x − 1 → 2 = 3 − 1 = 2 ✓

Check EQ 2: y = −2x + 8 → 2 = −2(3) + 8 = 2 ✓



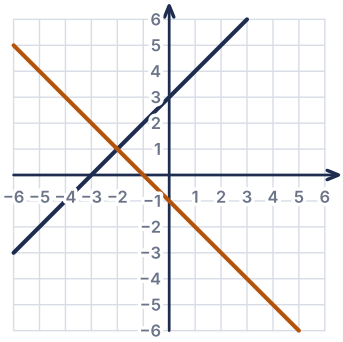
y = x − 1 y = −2x + 8

✏ Practice: Solving Systems by Graphing

1. Score 3 Where do the two lines cross? Write the ordered pair, like (2, 3).

Answer: (−2, 1)

How to do it: The lines cross at **(−2, 1)**. Check: −2 + 3 = 1 ✓ and −(−2) − 1 = 1 ✓.

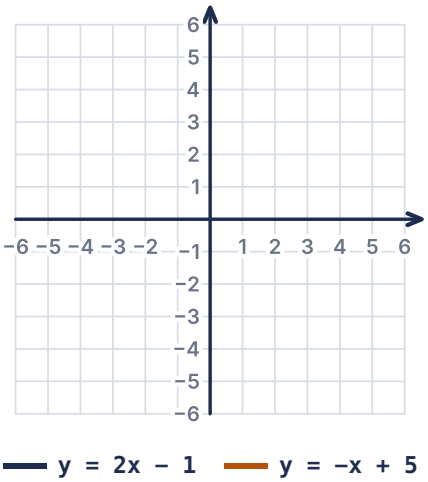


y = x + 3

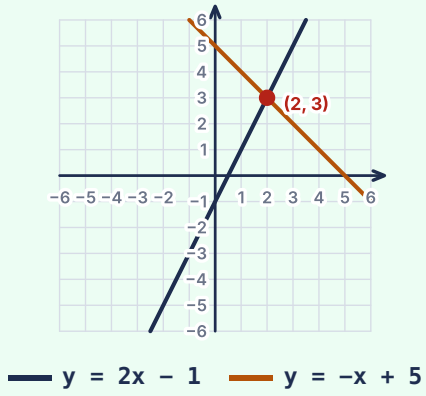
y = −x − 1

2. Score 4 Graph both lines on the grid. Where do they cross? Write the ordered pair.

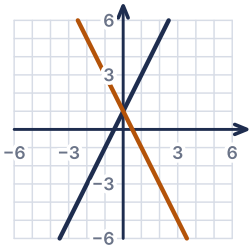
Answer: (2, 3)



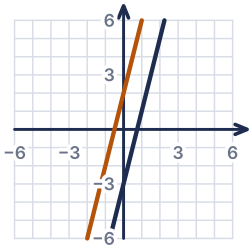
How to do it: y = 2x − 1: start at (0, −1), then up 2, right 1. y = −x + 5: start at (0, 5), then down 1, right 1. They cross at **(2, 3)**. Check: 2(2) − 1 = 3 ✓ and −2 + 5 = 3 ✓.



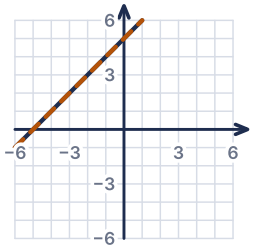
3. Score 5 Which system has **no solution**?



y = 2x + 1 and y = −2x + 1



y = 4x − 3 and y = 4x + 2



y = x + 5 and y = x + 5

- ○ y = 2x + 1 and y = −2x + 1
- ● **y = 4x − 3 and y = 4x + 2**
- ○ y = x + 5 and y = x + 5

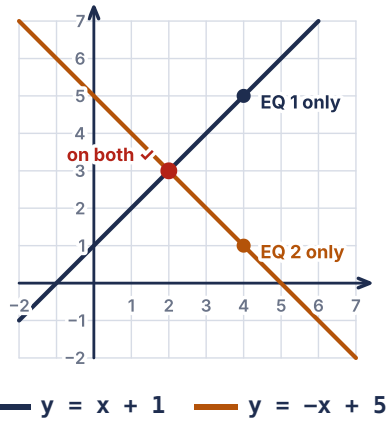
How to do it: First: slopes 2 and −2 are different → one solution. Second: slopes 4 and 4 are the same, y-intercepts −3 and 2 are different → parallel → **no solution**. Third: the same line twice → infinitely many.

3 What Does "Solution" Actually Mean?

The Core Definition

A **solution** to a system of equations is an ordered pair (x, y) that makes **every** equation in the system true at the same time.

> **TWO WAYS TO SAY IT**
>
> - **Graphically**: the solution is the point where the lines cross.
> - **Algebraically**: the solution is the (x, y) that makes both equations true.



Only the crossing point (2, 3) sits on **both** lines, so only it is a solution.

Testing Whether a Point Is a Solution

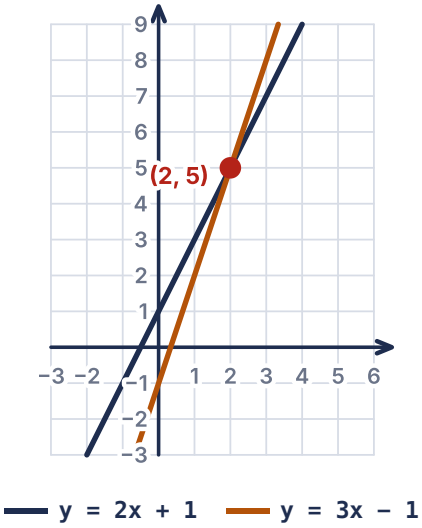
To test a candidate point (a, b), plug **x = a** and **y = b** into **each** equation. If the point makes *both* equations true, it's a solution. If it fails even one, it's **not** a solution.

EXAMPLE **Is (2, 5) a solution?**

EQ 1 $y = 2x + 1$

EQ 2 $y = 3x - 1$

1. **Test EQ 1** with x = 2, y = 5:
   $5 = 2(2) + 1 = 5$ ✓ True.
2. **Test EQ 2** with x = 2, y = 5:
   $5 = 3(2) - 1 = 5$ ✓ True.
3. Both equations are true, so **(2, 5) is a solution**.

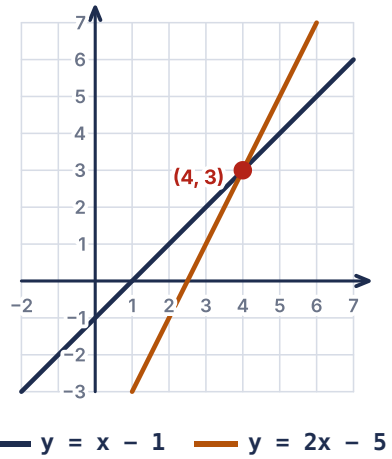


EXAMPLE **Is (4, 3) a solution?**

EQ 1 $y = x - 1$

EQ 2 $y = 2x - 5$

1. **Test EQ 1** with x = 4, y = 3:
   $3 = 4 - 1 = 3$ ✓ True.
2. **Test EQ 2** with x = 4, y = 3:
   $3 = 2(4) - 5 = 3$ ✓ True.
3. Both true, so **(4, 3) *is* a solution**. Even though it looked suspicious, the test confirms it.



TAKEAWAY

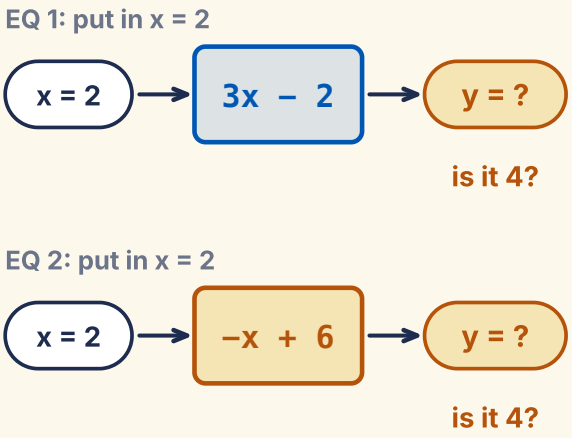
You cannot tell by eyeballing. Always test **both** equations.

▶ Try It · Test the Point

EQ 1 $y = 3x - 2$

EQ 2 $y = -x + 6$

Is **(2, 4)** a solution to this system?



Circle one: (YES) NO

Yes — (2, 4) is a solution.

EQ 1: $4 = 3(2) - 2 = 4$ ✓

EQ 2: $4 = -(2) + 6 = 4$ ✓

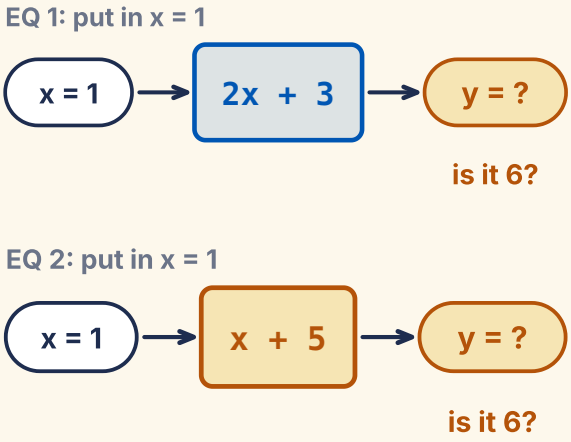
Both true.

▶ Try It · Test the Point (2)

EQ 1 $y = 2x + 3$

EQ 2 $y = x + 5$

Is **(1, 6)** a solution to this system?



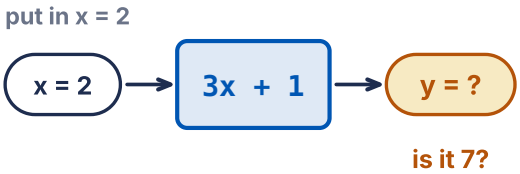
Circle one: YES **NO**

No — (1, 6) is not a solution.
EQ 1: 6 = 2(1) + 3 = 5. 6 ≠ 5 × (fails EQ 1)
EQ 2: 6 = 1 + 5 = 6 ✓
A point must satisfy *both* to be a solution. The real solution here is (2, 7).

✏ Practice: What "Solution" Means

1. Score 3 Does the point **(2, 7)** make **y = 3x + 1** true?

● **Yes**
○ No



How to do it: 3(2) + 1 = 6 + 1 = 7. That matches y = 7, so **yes** ✓.

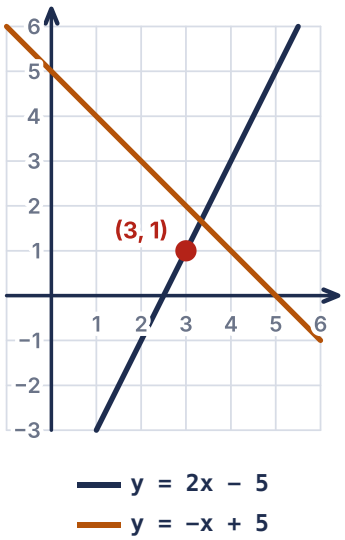
2. Score 4 Is **(3, 1)** a solution of this system?

> **EQ 1** $y = 2x - 5$
>
> **EQ 2** $y = -x + 5$

○ Yes

● **No**

> **How to do it:** EQ 1: 2(3) − 5 = 1 ✓ true. EQ 2: −(3) + 5 = 2, but y is 1 ×. It fails EQ 2, so **no**, (3, 1) is not a solution.

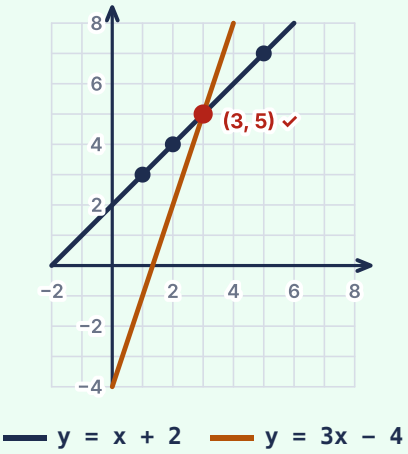


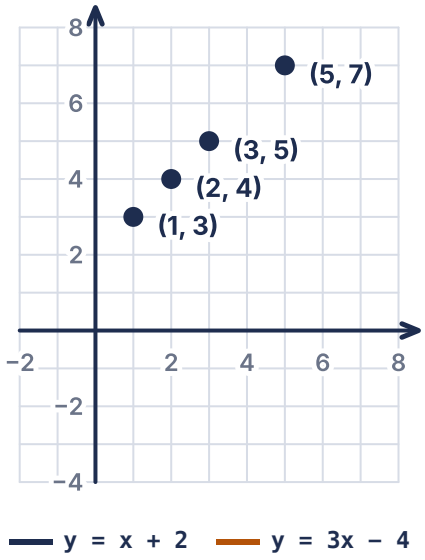
3. Score 5 Which point is the solution of this system?

> **EQ 1** $y = x + 2$
>
> **EQ 2** $y = 3x - 4$

- ○ (1, 3)
- ○ (2, 4)
- ● **(3, 5)**
- ○ (5, 7)

How to do it: All four points work in EQ 1. Now test EQ 2: $3(1) - 4 = -1$ ×, $3(2) - 4 = 2$ ×, $3(3) - 4 = 5$ ✓, $3(5) - 4 = 11$ ×. Only **(3, 5)** works in both.



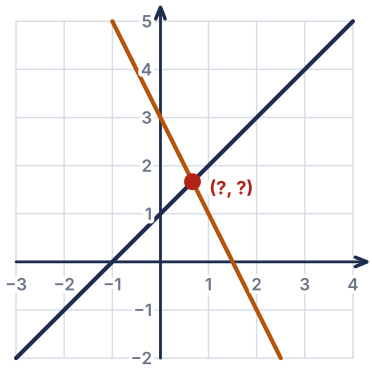


The 4 choices are plotted. Graph both lines to check.

4 Solving Systems by Substitution

Why Substitution?

Graphing is hard when the lines cross between grid lines. **Substitution** gives the exact answer.



y = x + 1 **y = −2x + 3**

These lines cross *between* grid lines. Reading the exact point is hard, so we use algebra.

The Steps

SUBSTITUTION PROCEDURE

1. Make sure one equation is solved for a variable (usually y = ...).
2. **Substitute** the expression into the *other* equation.
3. Solve that equation for x.
4. Plug x back in to find y.
5. Write the solution as an ordered pair (x, y) and check.

EXAMPLE 1 **Both equations in y = form**

EQ 1 $y = 2x + 3$

EQ 2 $y = -x + 6$

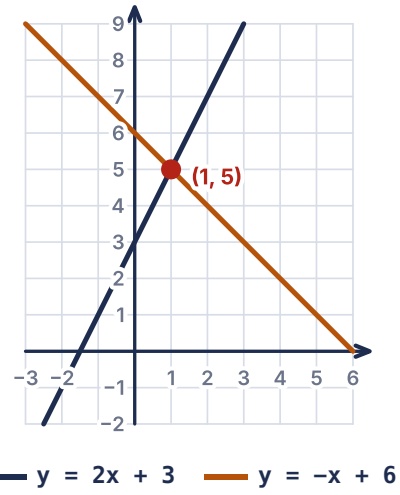
1. Both equations are solved for y. Since they both equal y, they must equal each other. Set them equal: $2x + 3 = -x + 6$
2. Solve for x. Add x to both sides: $3x + 3 = 6$. Subtract 3: $3x = 3$. Divide: $x = 1$.
3. Put your x into either equation to find y. Using EQ 1: $y = 2(1) + 3 = 5$.
4. **Solution: (1, 5)**
5. Check in EQ 2: $5 = -(1) + 6 = 5$ ✓
6. **Set them equal:**

1	$2x + 3 = -x + 6$ $+x \quad +x$	↓ Add x to both sides
2	$3x + 3 = 6$ $-3 \quad -3$	↓ Subtract 3
3	$\frac{3x}{3} = \frac{3}{3}$	↓ Divide
4	$x = 1$	

EQ 1 $y = 2x + 3$

EQ 2 $y = -x + 6$

so $2x + 3 = -x + 6$



EXAMPLE 2 **One equation in y = form, one not (Challenge)**

EQ 1 $y = 3x - 1$

EQ 2 $2x + y = 9$

1. EQ 1 is already solved for y. So $y = 3x - 1$.
2. Substitute $(3x - 1)$ in place of y in EQ 2: $2x + (3x - 1) = 9$
3. Solve for x. Combine like terms: $5x - 1 = 9$. Add 1: $5x = 10$. Divide: $x = 2$.
4. Put your x into EQ 1 to find y: $y = 3(2) - 1 = 5$.
5. **Solution: (2, 5)**
6. Check EQ 2: $2(2) + 5 = 9$ ✓
7. **Substitute (3x − 1) in place of y in EQ 2:**

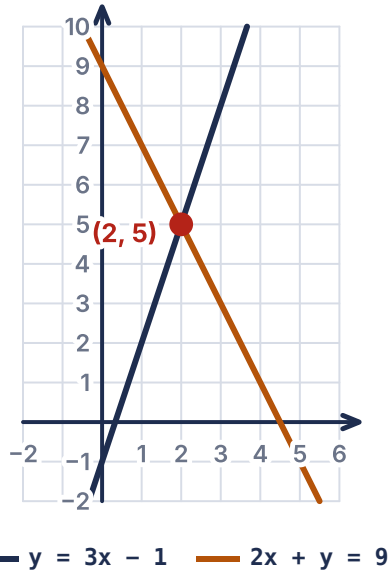
1	$2x + (3x - 1) = 9$	↓ Combine like terms
2	$5x - 1 = 9$ $+1 \quad +1$	↓ Add 1
3	$\frac{5x}{5} = \frac{10}{5}$	↓ Divide
4	$x = 2$	

EQ 1 $y = 3x - 1$

swap in

EQ 2 $2x + y = 9$

so $2x + (3x - 1) = 9$



EXAMPLE 3 **Quick rearrange required (Challenge)**

EQ 1 $x + y = 7$

EQ 2 $y = 2x + 1$

1. EQ 2 is already solved for y.
2. Substitute $(2x + 1)$ for y in EQ 1: $x + (2x + 1) = 7$
3. Solve: $3x + 1 = 7$ → $3x = 6$ → $x = 2$.
4. Put your x into EQ 2 to find y: $y = 2(2) + 1 = 5$.
5. **Solution: (2, 5)**
6. **Substitute (2x + 1) for y in EQ 1:**

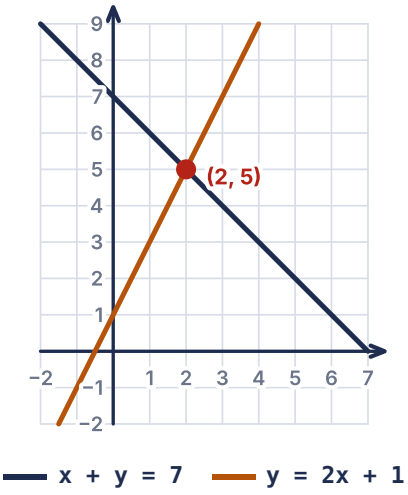
1	$x + (2x + 1) = 7$	↓ simplified
2	$3x + 1 = 7$ $-1 \quad -1$	↓ subtracted 1 on both sides
3	$\frac{3x}{3} = \frac{6}{3}$	↓ divided both sides by 3
4	$x = 2$	

EQ 2 $y = \boxed{2x + 1}$

swap in

EQ 1 $x + \boxed{y} = 7$

so $x + \boxed{(2x + 1)} = 7$



CHALLENGE NOTE

If *neither* equation is solved for a variable, pick the easier one and solve it first. If you see a coefficient of 1 on x or y, that equation is easiest.

▶ Try It · Substitution Practice 1

EQ 1 $y = x + 4$

EQ 2 $y = 3x - 2$

Solve the system using substitution. Write the solution as (x, y).

EQ 1 $y = x + 4$

EQ 2 $y = 3x - 2$

so $x + 4 = 3x - 2$

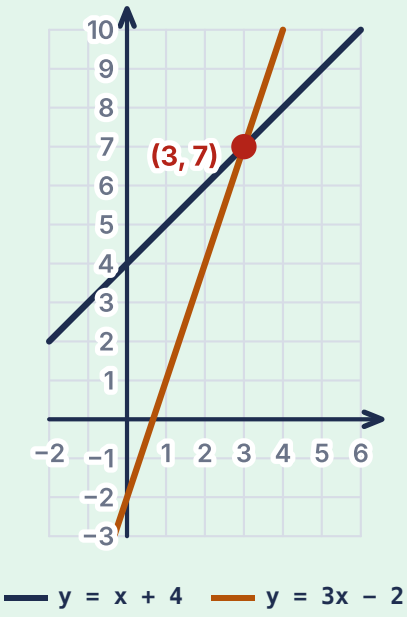
Solution: (3 , 7)

Solution: (3, 7)

Set equal: x + 4 = 3x − 2

4 + 2 = 3x − x → 6 = 2x → x = 3

y = 3 + 4 = 7



▶ Try It · Substitution Practice 2 Score 5

EQ 1 $y = 2x - 3$

EQ 2 $x + y = 9$

Solve using substitution. (EQ 1 is already done for you — sub into EQ 2.)

EQ 1 $y = \boxed{2x - 3}$

swap in

EQ 2 $x + \boxed{y} = 9$

so $x + \boxed{(2x - 3)} = 9$

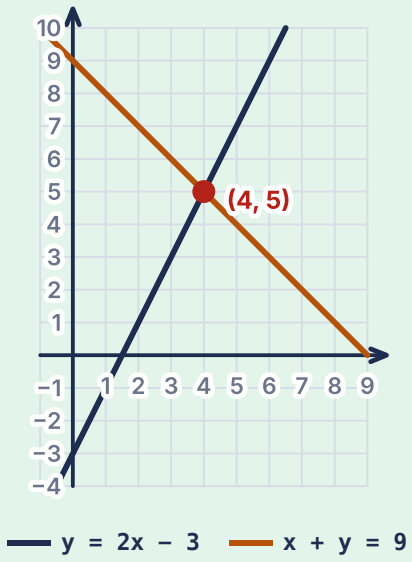
Solution: (4 , 5)

Solution: (4, 5)

Sub (2x − 3) for y in EQ 2: x + (2x − 3) = 9

3x − 3 = 9 → 3x = 12 → x = 4

y = 2(4) − 3 = 5



▶ Try It · Substitution Practice 3 (quick rearrange) Score 5

EQ 1 $2x + y = 10$

EQ 2 $y = x + 1$

EQ 2 is solved for y — sub (x + 1) for y into EQ 1.

EQ 2 $\quad y = \boxed{x + 1}$

swap in

EQ 1 $\quad 2x + \boxed{y} = 10$

so $\quad 2x + \boxed{(x + 1)} = 10$

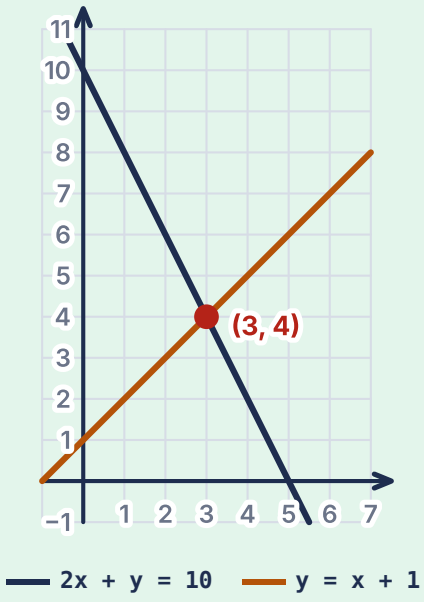
Solution: (3 , 4)

Solution: (3, 4)

Sub (x + 1) for y in EQ 1: 2x + (x + 1) = 10

3x + 1 = 10 → 3x = 9 → x = 3

y = 3 + 1 = 4



Graphing vs. Substitution — When to Use Which

GRAPHING

Good for seeing the answer. Hard when the answer is not a whole number.

SUBSTITUTION

Best when at least one equation is already solved for x or y (or is easy to solve). Always gives an exact answer — fractions and all.

✏ Practice: Solving Systems by Substitution

1. Score 5 Solve for x. Write just the x-value.

EQ 1 $y = 2x$

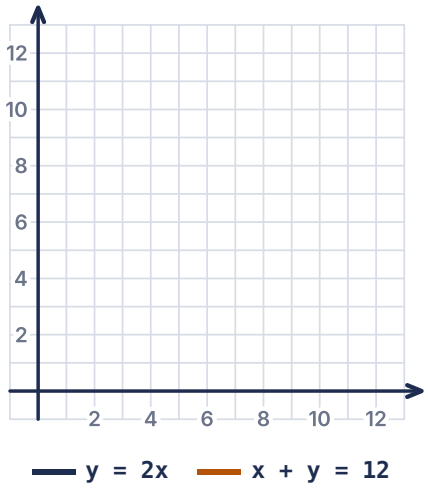
EQ 2 $x + y = 12$

Answer: x = 4

How to do it:

Put 2x in place of y:

1. $x + 2x = 12$ ↓ simplified
2. $3x = 12$



Check your answer: graph both lines here.

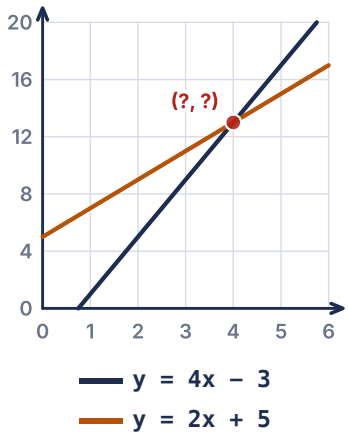
2. Score 4 Solve by substitution. Write the ordered pair (x, y).

> **EQ 1** $y = 4x - 3$
>
> **EQ 2** $y = 2x + 5$

Answer: (4, 13)

How to do it:

1. $4x - 3 = 2x + 5$
   $-2x \qquad -2x$
   ↓ subtracted 2x on both sides
2. $2x - 3 = 5$
   $+3 \qquad +3$
   ↓ added 3 on both sides
3. $\frac{2x}{2} = \frac{8}{2}$
   ↓ divided both sides by 2
4. $x = 4$



The lines cross between grid lines. Substitution gives the exact point.

3. Score 5 Gym A costs y = 5x + 10 dollars and Gym B costs y = 2x + 25 dollars for x classes. After how many classes do they cost the same?

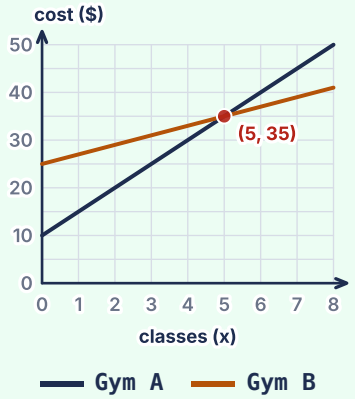
You can also finish the table. Look for the row where the costs match.

x	Gym A ($)	Gym B ($)
0	10	25
1	15	27
2	20	29
3		
4		
5		
6		

Answer: 5 classes

How to do it: 5x + 10 = 2x + 25 → 3x + 10 = 25 → 3x = 15 → x = **5**. After 5 classes both cost 5(5) + 10 = $35.

x	Gym A ($)	Gym B ($)
0	10	25
1	15	27
2	20	29
3	25	31
4	30	33
5	35	35
6	40	37



1. $5x + 10 = 2x + 25$ → subtracted $2x$ on both sides
   $-2x \qquad -2x$
2. $3x + 10 = 25$ → subtracted 10 on both sides
   $-10 \qquad -10$
3. $\frac{3x}{3} = \frac{15}{3}$ → divided both sides by 3
4. $x = 5$

5 Comprehensive Practice — All Sections

STANDARDS

NC.8.EE.8 — solve a system of two linear equations in slope-intercept form, and understand that the solution is the point where the graphs intersect. NC.8.F.4 — write a linear equation from a table. Parts A–C practice graphing; Part D mixes every section.

A. Read the Graph

Both lines are already drawn. Find the point where they cross, then write that ordered pair.

1 · Read the intersection Score 3

EQ 1 $y = x + 2$

EQ 2 $y = -2x + 8$

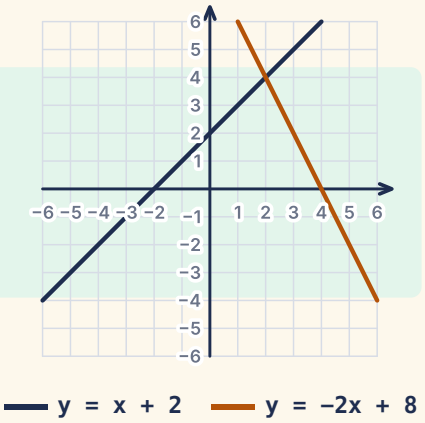
Solution: (2 , 4)

Solution: (2, 4)

EQ 1: 4 = (2) + 2 = 4 ✓

EQ 2: 4 = −2(2) + 8 = 4 ✓

Both true.



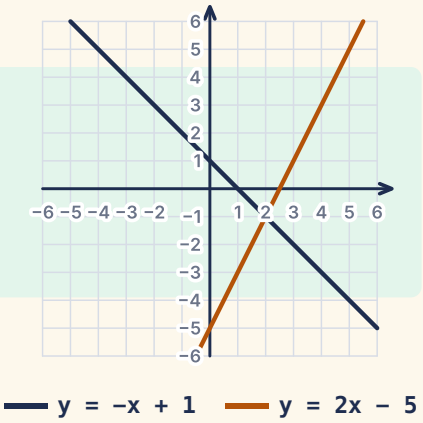
▶ **2 · Read the intersection** Score 4

EQ 1 $y = -x + 1$

EQ 2 $y = 2x - 5$

Solution: (2 , −1)

Solution: (2, -1)
EQ 1: −1 = −(2) + 1 = −1 ✓
EQ 2: −1 = 2(2) − 5 = −1 ✓
Both true.



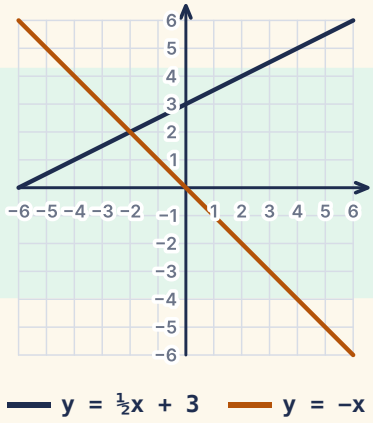
▶ **3 · Read the intersection** Score 5

EQ 1 $y = \frac{1}{2}x + 3$

EQ 2 $y = -x$

Solution: (−2 , 2)

Solution: (-2, 2)
EQ 1: 2 = ½(−2) + 3 = 2 ✓
EQ 2: 2 = −(−2) = 2 ✓
Both true.



B. Graph It Yourself

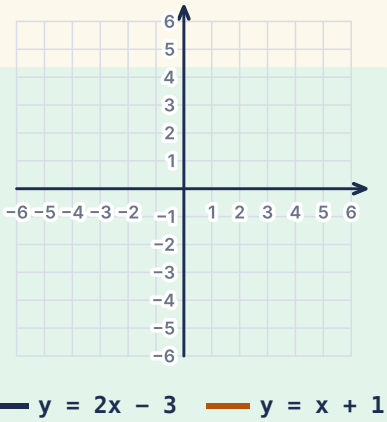
The grid is empty on purpose. Graph each line from its slope and y-intercept, read the intersection, then write the point where they cross.

1 · Graph and solve Score 4

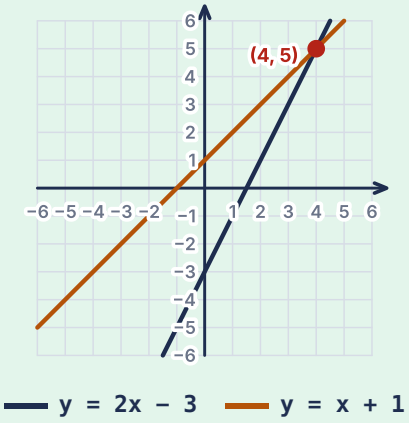
EQ 1 $y = 2x - 3$

EQ 2 $y = x + 1$

Solution: (4 , 5)



Solution: (4, 5)



EQ 1: $5 = 2(4) - 3 = 5$ ✓

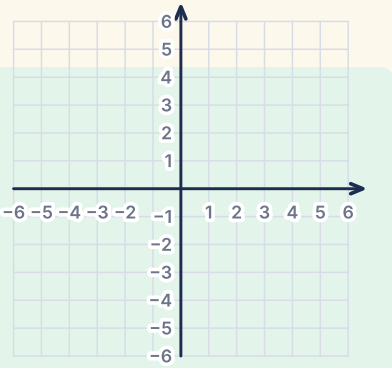
EQ 2: $5 = (4) + 1 = 5$ ✓

2 · Graph and solve Score 4

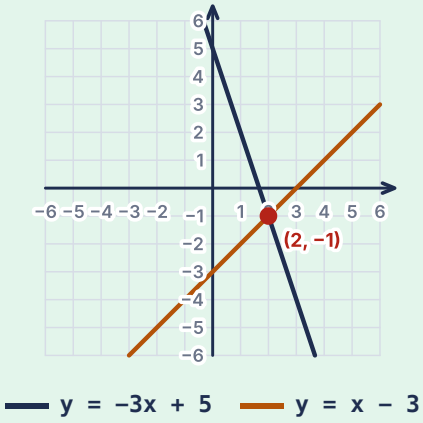
EQ 1 $y = -3x + 5$

EQ 2 $y = x - 3$

Solution: (2 , −1)



Solution: (2, -1)



EQ 1: $-1 = -3(2) + 5 = -1$ ✓

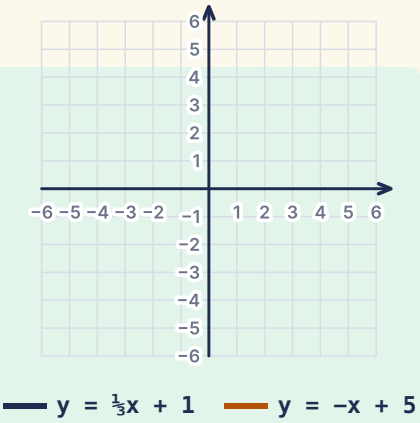
EQ 2: $-1 = (2) - 3 = -1$ ✓

3 · Graph and solve Score 5

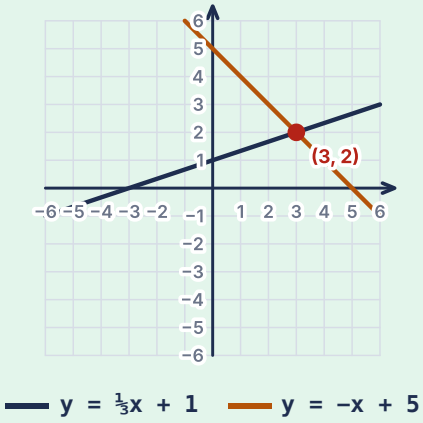
EQ 1 $y = \frac{1}{3}x + 1$

EQ 2 $y = -x + 5$

Solution: (3 , 2)



Solution: (3, 2)



EQ 1: $2 = \frac{1}{3}(3) + 1 = 2$ ✓

EQ 2: $2 = -(3) + 5 = 2$ ✓

C. One, None, or Infinitely Many?

Compare the slopes and y-intercepts *before* you look at the picture.

▶ **1 · How many solutions?** Score 3

EQ 1 $y = 2x + 1$

EQ 2 $y = 2x - 4$

EQ 1 $y = 2x + 1$

EQ 2 $y = 2x - 4$

slopes 2 and 2

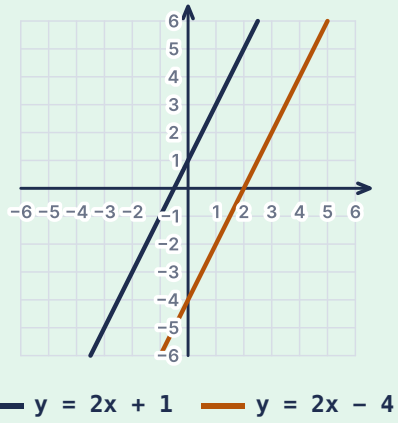
y-ints 1 and −4

Same or different? Decide.

Answer: No solution

No solution

Same slope (2 and 2), different y-intercepts (1 and −4). The lines are parallel, so they never meet.



▶ **2 · How many solutions?** Score 4

EQ 1 $y = 3x - 2$

EQ 2 $y = -x + 2$

EQ 1 $y = \boxed{3}x \boxed{-2}$

EQ 2 $y = \boxed{-1}x \boxed{+2}$

slopes **3 and −1**

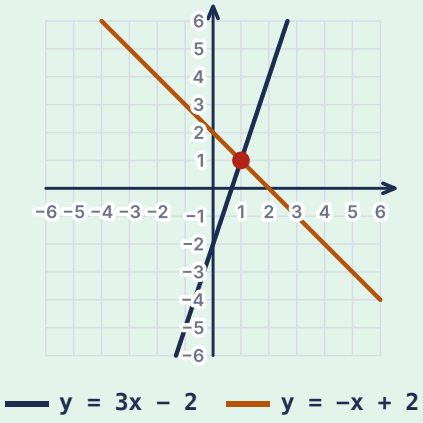
y-ints **−2 and 2**

Same or different? Decide.

Answer: One solution — (1, 1)

One solution — (1, 1)

Different slopes (3 and −1), so the lines cross exactly once. Check: 3(1) − 2 = 1 and −(1) + 2 = 1. ✓



3 · How many solutions? Score 4

EQ 1 $y = -\frac{1}{2}x + 3$

EQ 2 $y = -\frac{1}{2}x + 3$

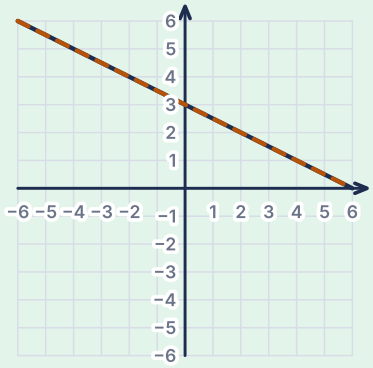
EQ 1	$y = -\frac{1}{2}x + 3$
EQ 2	$y = -\frac{1}{2}x + 3$
slopes	$-\frac{1}{2}$ and $-\frac{1}{2}$
y-ints	3 and 3

Same or different? Decide.

Answer: Infinitely many solutions

Infinitely many solutions

Same slope AND same y-intercept — this is one line written twice, so every point on it satisfies both equations.



Same line twice

$y = -\frac{1}{2}x + 3$ $y = -\frac{1}{2}x + 3$

D. Mixed Review — All Sections

✏ Comprehensive Practice Tables, graphing, solutions, substitution

1. Score 3 In **y = 5x − 2**, what is b, the y-intercept? Write just the number.

Answer: b = −2

How to do it: y = mx + b. Here m = 5 and b = **−2**.

form $y = mx + b$

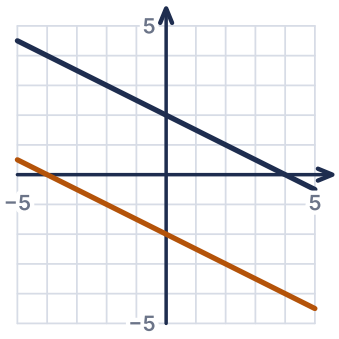
yours $y = 5x - 2$

Line them up. What is in the b spot?

2. Score 3 Two lines are parallel. How many solutions does the system have?

- ○ One solution
- ● **No solution**
- ○ Infinitely many solutions

How to do it: Parallel lines never cross, so there is **no solution**.

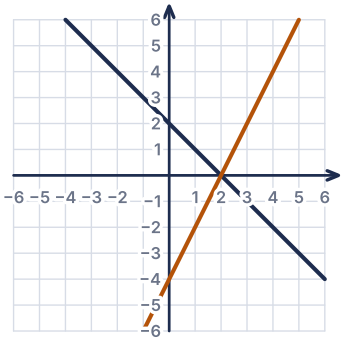


Two parallel lines

3. Score 3 Where do the two lines cross? Write the ordered pair.

Answer: (2, 0)

How to do it: The lines cross on the x-axis at **(2, 0)**. Check: −2 + 2 = 0 ✓ and 2(2) − 4 = 0 ✓.



y = −x + 2

y = 2x − 4

4. Score 4 Write the equation for this table in the form y = mx + b.

Answer: y = 3x + 1

x	y
0	1
2	7
4	13
6	19

How to do it: When x goes up 2, y goes up 6, so m = 6 ÷ 2 = 3. When x = 0, y = 1, so b = 1. Equation: **y = 3x + 1**.

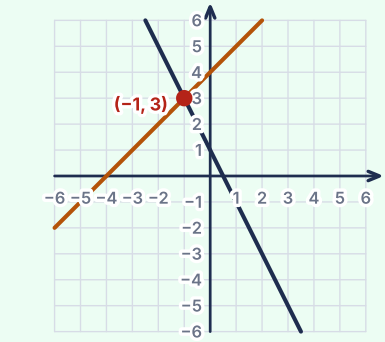
5. Score 4 Is **(−1, 3)** a solution of this system?

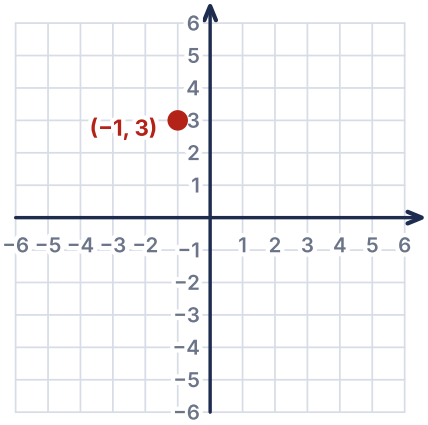
EQ 1 **y = −2x + 1**

EQ 2 **y = x + 4**

- ● **Yes**
- ○ No

How to do it: EQ 1: −2(−1) + 1 = 2 + 1 = 3 ✓. EQ 2: −1 + 4 = 3 ✓. Both true, so **yes**.





Graph both lines. Do both go through the point?

6. Score 4 Solve for x. Write just the x-value.

EQ 1 $y = 3x$

EQ 2 $x + y = 8$

Answer: x = 2

How to do it:

1. $x + 3x = 8$ ↓ simplified
2. $4x = 8$

EQ 1 $y = 3x$

swap in

EQ 2 $x + y = 8$

so $x + (3x) = 8$

7. Score 4 Solve by substitution. Write the ordered pair (x, y).

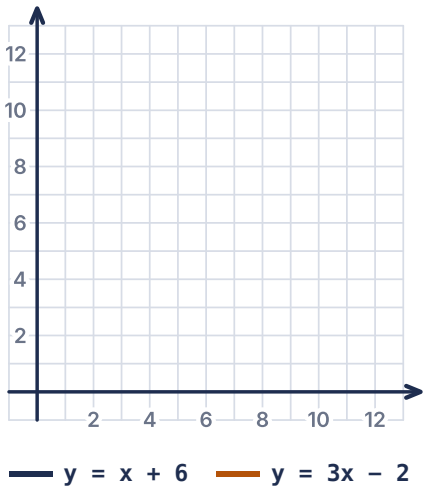
EQ 1 $y = x + 6$

EQ 2 $y = 3x - 2$

Answer: (4, 10)

How to do it:

1. $x + 6 = 3x - 2$ ↓ subtracted $x - 2$ on both sides
   $-x + 2 \quad -x + 2$
2. $6 + 2 = 3x - x$ ↓ simplified
3. $8 = 2x$

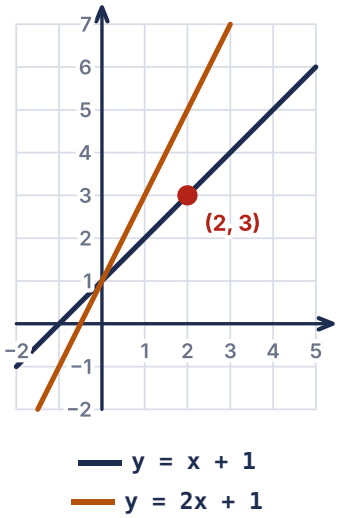


Check your answer: graph both lines here.

8. Score 5 Kai says (2, 3) is a solution of y = x + 1 and y = 2x + 1, because it works in the first equation. Is Kai right?

- ○ Yes, one true equation is enough
- ● **No, it must make both equations true**

How to do it: EQ 1: 2 + 1 = 3 ✓. EQ 2: 2(2) + 1 = 5, not 3 ×. A solution must work in **both** equations, so Kai is wrong.

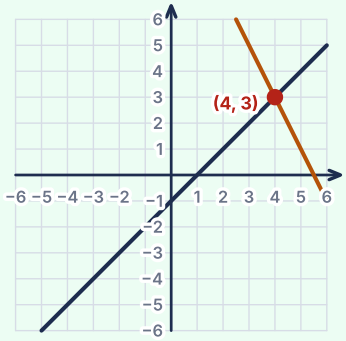


9. Score 5 Neither equation is solved for y. Solve the system and write the ordered pair.

> **EQ 1** $x - y = 1$
>
> **EQ 2** $2x + y = 11$

Answer: (4, 3)

How to do it: Solve EQ 1 for y: y = x − 1. Substitute: 2x + (x − 1) = 11 → 3x − 1 = 11 → 3x = 12 → x = 4. y = 4 − 1 = 3. Solution: **(4, 3)**. Check EQ 2: 2(4) + 3 = 11 ✓

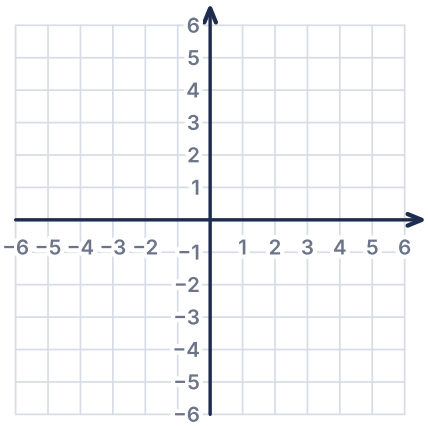


x − y = 1 (y = x − 1)

2x + y = 11 (y = −2x + 11)

Substitute:

1	$2x + (x - 1) = 11$	↓ simplified
2	$3x - 1 = 11$ $+1 \quad +1$	↓ added 1 on both sides
3	$3x = 12$	



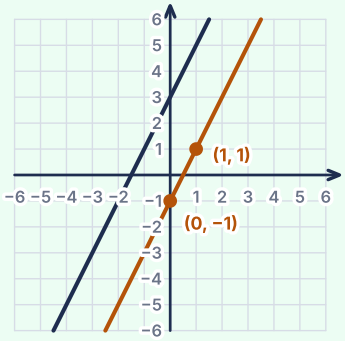
x − y = 1 2x + y = 11

Check your answer: graph both lines here.

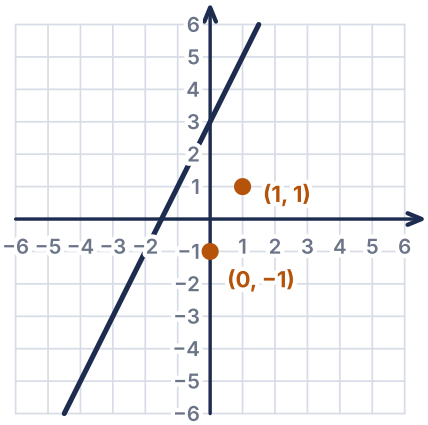
10. Score 5 Line A is y = 2x + 3. Line B goes through (0, −1) and (1, 1). How many solutions does the system have?

- ○ One solution
- ● **No solution**
- ○ Infinitely many solutions

How to do it: Line B: slope = (1 − (−1)) ÷ (1 − 0) = 2, and b = −1, so y = 2x − 1. Same slope as Line A (2), different y-intercepts (3 and −1) → parallel → **no solution**.



Line A: y = 2x + 3 Line B: y = 2x − 1



Line A: y = 2x + 3
Line B: through the 2 points

Draw Line B through the two gold points.